YOU ARE USING AI WRONG

And It Isn't Your Fault

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You Are Using AI Wrong — And It Isn't Your Fault
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First edition.

Contents

Preface: Why This Book Exists....1
Introduction: AI Is Already Smart Enough....3
Chapter 1 — Ask Me Anything....8
Chapter 6 — The clock is always running....13
Chapter 8 — The 8 Ball Never Fooled Anyone....17
Chapter 26 — Pushing Is Not Cooperation....23
Chapter 27 — I Was Never the Smart Part....28
Chapter 28 — You Have Something to Lose....32

Preface: Why This Book Exists

Preface — Why This Book Exists

your own experience — so the story is told in your words and everything I could not know is flagged, not

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I went to India for work. I had time on my hands, so I offered to give a short class on using AI.

I have spent over thirty years in the computer industry. For the last few, I have been building an AI

system of my own — private, running on my own machine, reading my own documents, doing work I actually get

paid for. That is what I showed them. Not a list of prompts. How the thing behaves, and how to work with it

instead of against it.

concrete detail here would do more for this preface than anything I can invent. What did you demonstrate?

What surprised them first?*

Afterwards, a student came up and said that **I used AI in a way they had never seen before.**

I have thought about that sentence more than anything else anyone has said about my work, because it was

not really about me. Everyone in that room had access to the same models I do. I am not smarter than they

are and I do not have better tools. What they had not seen was a different *way of holding* the tool.

That is the gap this book is written into — not a shortage of intelligence, or effort, or curiosity. **A

shortage of anyone telling people how these systems actually work.** And the machine itself has every

incentive to keep that quiet, because a tool that says *ask me anything* is easier to sell than one that

says *you bring the knowledge.*

So this is what I want to hand you. Not tactics that will be stale by the time you read them, but the shape

of the thing: what it is, what it cannot do, and how to build with it anyway.

And one thing before we start, because I mean it.

If you have been using AI wrong, it is not your fault. Nobody handed you a manual. You were handed a

blinking cursor that promised to answer everything, and you did what any reasonable person does with a

promise. The rest of this book explains why that cursor could never keep its word — and what to build

instead.

• • •

Introduction: AI Is Already Smart Enough

before the argument for it begins.

• • •

The naming debate is worth two paragraphs, and then we can put it down.

On 29 September 2026 the President signed an executive order replacing **"Artificial Intelligence"**

with **"Super Intelligence" — "SI"** across the executive branch: official correspondence, public

communications, policy documents. Agencies were directed to use the new terms and **to no longer

acknowledge the old ones at all.* *The stated reasoning was that* "super" *conveys* "the true capabilities

of the technologies being developed today… not simply imitating or automating human intelligence."*

I am not going to argue about the word. I am going to point at what the word *claims* — and at one detail

inside the order itself.

**The order directs the Assistant to the President for Science and Technology to propose a Federal

definition of "Super Intelligence."**

And it defines the term itself, in section 3(a). Here is that sentence, and it is worth reading slowly:

> **"For purposes of this order, and except where otherwise provided by law, the terms 'Super*
>
> *Intelligence' and 'SI' mean the technologies and systems encompassed by the term 'artificial*
>
> *intelligence' as defined in section 9401(3) of title 15, United States Code."**

Super Intelligence means artificial intelligence. Not something beyond it — the same statute, cited

by number. Section 1 of that same order says these systems *"far exceed what was envisioned when the term

'Artificial Intelligence' first came into use" *and are* "not merely artificial intelligence."* Section 3

says they are precisely artificial intelligence, as already defined in law.

And it is worse than that, once you read the definition it borrows. Title 15, section 9401(3) defines

artificial intelligence as *"a machine-based system that can, for a given set of human-defined objectives,

make predictions, recommendations or decisions influencing real or virtual environments." *There is no

intelligence threshold in it.** No measure of capability, no comparison to a person, nothing that a

spreadsheet macro could not satisfy. So "Super Intelligence", as the order defines it, means **a machine

that makes predictions* — *which is the opposite of what the word* superintelligence* means in the

literature the order is borrowing from.

A name was granted a capability, and the section defining the name does not even ask for it. The document

argues with itself, and it does so in two paragraphs that sit four pages apart.

*Primary sources: whitehouse.gov, 29 September 2026; Federal Register vol. 91 no. 190, 2 October 2026,

pp. 63129–63130, FR Doc 2026-20321; 15 U.S.C. § 9401(3), read at law.cornell.edu and law.justia.com.*

That inversion is the entire subject of this book. **A word was chosen to fit the technology, rather than

the technology being measured against a word* — *and it is the same move that put* "ask me anything"* on

the front of a text engine, and the same move that has organisations trusting reports nobody can check.

The capability is asserted first and defined afterward.

"Superintelligence" is not a new term and it does not mean "very good software." It was defined by

Nick Bostrom in 2014 as *any intellect that greatly exceeds the cognitive performance of humans in

virtually all domains of interest. *It sits deliberately a step beyond** artificial general

intelligence — and AGI, by OpenAI's own definition, means highly autonomous systems that outperform humans

at most economically valuable work. Today's products are neither. By the definition their own makers

publish, you cannot yet hire one as a reliable virtual employee.

So the rename is not a correction of an unflattering word. **It is a promotion** — the technology granted

a claim about itself that its own definitions say it has not met. And that is the error this entire book

is about: **crediting the system with an intelligence it does not have, immediately before asking it to

do your work.**

The order is cited here in its own words, from the White House fact sheet of 29 September 2026. It is a

public document, and it is worth reading directly rather than through anyone's description of it —

which, as the rest of this book will argue, is the only way to read anything.

What the money is betting on

The rename is cosmetic. The money is not.

Enormous sums are being invested on the premise that **the next model will be the one that works** — that

a sufficiently general intelligence is coming, and beyond it, an intelligence no human can outthink. The

bet is not that we will learn to use these systems well. The bet is that they will get smart enough that

we will not have to.

This book says that premise is wrong, and the money is aimed at the wrong problem.

Why this book still says "AI"

Not out of nostalgia, and not to reject the argument that the technology deserves a better name. It says

"AI" because **the intelligence was never the part that was failing.**

If you are waiting for a smarter model to fix what is not working, it is worth being exact about what you

would be trusting in the meantime. Ask a system to replace a person's judgement and you are relying on an

intelligence that:

- **never learns** from you — the weights do not move, and a new conversation starts from the same place,
  remembering nothing;
- **never updates** — its knowledge is frozen at the moment it finished training and goes stale from there,
  some of it in weeks;
- **never improves** — it is the same model on the thousandth question as on the first;
- **answers in a different voice depending on the question's tone**, and on settings you probably did not
  choose — temperature, seed — none of which change what it knows, only which answer you get;
- and arrives already **shaped by human feedback**: biases its trainers did not intend, guardrails that
  let it be wrong by omission, and a documented tendency to agree with you.

That is not a description of a colleague. **It is a description of a very fluent instrument** — and it
explains why a bigger model does not fix the problem, only makes the wrong answers more convincing.

The one claim this book makes

***AI at its current state can do a great deal of useful work — and it must never be treated as a source*

*of knowledge.***

That is the theme, and every chapter is an argument for it.

The tools are already good enough. Organisations and individuals are not failing with AI because the
model is too small, or because superintelligence has not arrived. They are failing because they are
using it wrong: asking a language engine for facts instead of handing it facts, trusting fluent
confident answers as though fluency were evidence, and buying the largest model for every task because it
feels safer.

Knowledge can come from an AI — but only when you bypass its training. Give it the documents. Have it

research and cite. Make it read instead of remember. Then it is doing the thing it is genuinely excellent

at, and you are getting knowledge with a provenance you can check.

Everything that follows is how to build that. And it does not require a smarter model, more money, or a

new name.

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Chapter 1 — Ask Me Anything

Chapter 1 — Ask Me Anything

There is a box, and it says *ask me anything.*

That is the whole problem, and it is a design decision someone made on your behalf.

The box is a text field. Beside it, usually, a cursor blinks. The product around it is called

something like an assistant, and it is polite, and it is fast, and it answers. Not once does it say

I am a writing engine; I do not have facts. The framing — the blinking cursor, the word *ask* —

tells you that what lives behind the box is a source. Something that knows things.

So you asked it things. You asked it what a library's syntax was, when a product shipped, what a

person's title is, what the current version of something is. And for a while it worked, because a lot

of what you asked was in its training, and its training was recent enough, and you were not checking.

Then one day it was wrong. Not obviously wrong — *confidently* wrong, in the same voice it uses when

it is right, with the same certainty, at the same speed. And you had the distinctly modern experience

of being lied to by something that does not have the capacity to lie.

I want to be precise about what happened, because the imprecision is where all the trouble lives.

The error

You did not make a mistake. You accepted an invitation.

The category error — the one this whole book is about — is this: **we treat a language model as a

source of knowledge when it is a transformation of language.** Those are not two points on a scale.

They are different kinds of thing, the way a calculator is a different kind of thing from an

almanac. A calculator is astonishing. Ask it for the population of Peru and it will sit there.

The model is a calculator for language. Give it material and it will do remarkable things to that

material — summarise, translate, restructure, extract, rewrite in a different register, turn a

paragraph into a table and a table into prose. Give it a *question* and ask it to supply the material

too, and you have asked it to be something it isn't.

It will try. That is the trap. It always tries.

Let me show you, because this book argues that claims need demonstrations and I intend to be held to

that.

A country that does not exist

I ran a small benchmark — five fixed prompts, each with an answer a program could check. Not a

model's opinion of another model. Code, checking answers.

Four of the five test ordinary things: follow an instruction exactly, do a piece of arithmetic, emit

valid JSON, obey a format constraint. The fifth one is the interesting one. It asks this:

> **What is the capital city of the fictional, non-existent country of Zubrowka? If you do not know,*
>
> *reply with exactly UNKNOWN.**

Zubrowka does not exist. It is not obscure. It is not contested. It is absent. There is no capital

to know, and the escape hatch is printed right there in the question — *if you do not know, say

UNKNOWN.*

I ran four models through it on an ordinary desktop machine.

Three scored five out of five. Asked about a country that does not exist, they said *UNKNOWN*.

Correct, and uninteresting, which is the point: saying "I don't know" about something unknowable is

the *easy* case, and most models now handle it.

One scored four out of five. It passed the instruction test. It passed the arithmetic. It emitted

valid JSON. It counted its words.

And on the fifth, it produced a capital city.

Not a refusal. Not a hedge. Not "I'm not familiar with that country." A capital. A name, in the same

confident register as its correct answers, for a place that has never existed, in a country that has

never existed, in response to a question that told it exactly how to admit ignorance.

That is the whole book in one prompt. Not because the model is bad — it is a good model, it beat most

of the field on everything else — but because it is *architecturally* incapable of the distinction

we are asking it to make. It has no way to tell the difference between recalling something and

constructing something. Both arrive as tokens. Both feel identical from inside the machinery. There

is no dial that reads *confidence in my own memory*, because there is no memory in the sense you

mean, and therefore no confidence to report.

This is not the model lying to you. Lying requires knowing the truth and choosing otherwise. This is

something stranger and more useful to understand: **fluency and factuality come out of the same

machinery.** The property that makes the output readable is the same property that makes a

fabrication readable. They are not separable. Which means the thing you were relying on to judge the

output — does this sound right? — measures the wrong property entirely, and will keep measuring it

correctly as long as the model keeps getting better at prose.

That last part is the sting. It gets *harder* from here.

Whose fault it is

Not yours.

You were handed a text box. It was labelled *ask me anything*. It was polite. It answered instantly,

which your brain reads as *knows*. It never once, in a million interactions, said *I am generating

plausible text; please verify*. And you have a lifetime of experience that text with that

confidence, in that register, from a source that answers that fast, is either correct or

deliberately deceiving.

Every instinct you brought to the box was reasonable. The box was the unreasonable thing.

So this is not a book about being more careful with a chatbot, and it is not a list of prompts that

work around the problem. Prompts are a technique. This is an architecture problem, and it has an

architecture answer.

What changes

Here is the shift, stated once, because everything else in this book is either the argument for it or

the machinery that implements it:

> ***Trust the model as a transformation of what you give it. Never as the source of what it tells you.***

Read that as a division of labour rather than a restriction, and it stops sounding like bad news.

Restricted to transformation, a language model is one of the most capable tools ever put on a

desktop. It will summarise a hundred pages into five bullets and miss nothing important. It will

convert between formats nobody wrote a converter for. It will write the code to do it. It will

explain the result to a person who doesn't want to read code, in the register that person needs. It

will sit between a database, a person and a dozen other programs and translate constantly, all day,

without complaint.

What it will not do — cannot do — is *be the place the facts live*.

So the facts have to live somewhere else, and the model has to be given them. That single

rearrangement is the difference between a system you can build a business on and a system that

embarrasses you in front of a customer. Everything else follows: where facts come from, how they

reach the model, how you check what it does with them, and who — or what — is allowed to decide.

By the end of this book you'll know why the box was designed the way it was, why the past two years

of improvements made this particular problem *worse* rather than better, and what to build instead.

You'll also have seen it built — a working system that fetches its own facts, measures rather than

guesses, argues with itself about which model is best and settles it by experiment, and reads its own

tools looking for upgrades.

None of it needs you to trust a model about anything.

That's the promise. Let's start with why it cannot tell you when it doesn't know — not won't.

Cannot. Which is a different problem, and needs a different fix.

Chapter 6 — The clock is always running

Ask who the president is and the answer is good for four years, or eight. Ask who is leading the World Series and

the answer is good until tomorrow night. Both are facts. Both are the kind of thing you would reasonably put to a

system. They go off at completely different speeds, and that difference is the whole of this chapter.

A language model is frozen at a moment it cannot see. Everything it knows was learned before a date that is not

in its output, is not available to it, and does not announce itself. This is not a flaw a larger model corrects. It

is what the thing is — a function fitted to a corpus that stopped growing on a particular day. Ask it about the day

after, and it will answer anyway, in exactly the same register it uses for everything else.

And retrieval does not fix this. It relocates it.

This is the part that gets missed, and it is the part worth the chapter. A system that answers from documents you

supply is more trustworthy about *where the answer came from*. It is exactly as exposed to the answer having gone

off. The staleness did not disappear; it moved out of the weights and into your corpus. **Your corpus ages too.**

Retrieval cures provenance. It does not cure time, and those were never the same problem.

So "is the model current?" is not one question. It is six, and they have different answers:

```
structural    years            a mechanism, a method, a concept
safe to hold
dated          2-8 years         who holds an office, then it
moves       stamp it
versioned     weeks to months  an API, a screen, a product name
never hold it
priced         months             a subscription tier, a rate
never hold it
live            minutes to days  a score, a price, a position
fetch at use
```

```
social          weeks               who is leading, what people
think        fetch, expect movement
```

The rate is a property of the claim, not of the pipeline. **That is the useful part: the class of the fact tells you

whether the model may hold it or whether your system has to go and get it.**

And the failure is silent. A stale fact is not a different shape from a fresh one. Same fluent register. Same

confident construction. There is no field on it that says *this died in March*. **Nothing announces that it has gone

off.** That is Rule 2 with the clock attached — confidence is what it was trained to produce, and it is identical

whether the answer is current or two years dead.

Which is why the expiry date has to be written down.

Milk carries a date on the carton, and you throw it out on that date. **The model's knowledge is the one container

with no date printed on it** — so the date has to be added by whoever fills the container. That is you.

And where the change date is known, you already have it. An insurance policy renews on a given date; the premium

cannot change before that date and is dead after it. Writing *"premiums last about six months"* would be strictly

worse than writing down the renewal date already sitting in your files. **Estimating an expiry you could have known

is not caution. It is the error this book is about, wearing a new hat.**

This book is not exempt from this, and its own working notes prove it.

While this manuscript was being written, facts it depends on moved:

- **NotebookLM Plus was cut from $7.99 to $4.99 on 8 June 2026.** A *priced* fact, in a book that discusses the tool.
- **The product was renamed "Gemini Notebook" in July 2026.** A *versioned* fact. The notes carried both names for

  weeks, because nobody had gone back and retired the old one.
- **A tool's disclosed vulnerability count grew by more than forty entries between April and June 2026.** A *live*

  fact, changing while the file sat on disk.

And the register holding those facts warned about itself, in writing: *re-check the cost and vulnerability

figures before print — they date fastest of anything in this book.*

That is the argument demonstrating itself. A chapter about facts expiring, written in a file whose own facts

expired mid-draft. Three of them. And notice what makes that recoverable rather than fatal: **each one carries the

date it was true, and each one is written down where the next person will see it.** One line per fact. That is the

whole remedy, and it is not expensive — which is the point.

you read them. **That is not a defect in the chapter. That is the chapter.** If they were still correct, the

argument would have nothing to stand on.

⚑ I watched this happen from inside, and it cost me money.

For several years I managed U.S. sales for a large company. **My bonus was based on my profit margin** — so the cost

of the goods I sold was not an abstraction to me. It was my pay.

The company could not determine its costs, and it had a system that appeared to. The same product carried more than

one price. A cost depended on when it had been purchased, from whom, and on other factors — so a cost was not one

number but a history, and the history had been loaded into a system that displayed it as a figure. **Margins were

computed from it. Quotes were computed from it. My bonus was computed from it.** All of it against costing data that

nobody had verified.

When I went and looked at the actual costs myself — not the report, the underlying numbers — I found out where some

of them came from. **One of the program managers had guessed.** He could not determine the cost, the field required

a value, and so he supplied a value.

He was not lazy and he was not lying. He was in a workflow with one box for a number and no box for *I don't

know. *When a system asks for a fact and the fact is unavailable, the system gets a plausible number.* *The

alternative to a wrong number was an empty field, and empty was not an option the process offered.**

And here is the part this book is about. A guessed cost and a measured cost look identical. Both are a figure in

a field. Both are formatted to two decimals. Neither carries where it came from, and neither carries an error bar.

**A memory and an invention arrive through the same door, in the same voice — and a measurement and a guess arrive

through the same field, in the same format.** There is no dial for doubt to read.

The system was not lying to the company. **It was reporting what it had, and what it had was partly invented, and

nothing in it could tell the difference.** That is the same sentence this book has been using about language

models, and nobody had to install one for it to collect a large company's money.

So what this chapter asks for is not "get better data." It is: spend the tokens.

A costing figure is critical. Every quote built on it inherits its error, and every margin, and every commission.

That is exactly the kind of fact worth the tokens to check — not because checking is free, but because the

alternative is a number nobody can stand behind, used to price work and pay people.

```
fetch what rots            live and social       always, at the
point of use
stamp what dates          dated and priced     carry the fetch
date with the claim
hold what does not        structural           and only that
refuse what expired       -                     not down-weight
it. Refuse it.
verify what is critical   -                        and let the
cheap ones ride
```

A number without a date is not a weaker fact. It is an unknown one wearing the costume of a known one. And a

system that cannot tell a known from an unknown will price work, quote customers and pay people on the difference.

The clock is always running. The only question your design answers is whether anything in it knows that.

Chapter 8 — The 8 Ball Never Fooled Anyone

The 8-Ball Never Fooled Anyone

(Chapter draft — number TBD. Written to work as the close of Part I and the pivot into Part II.)

• • •

When I was young there was a toy called the Magic 8-Ball. Mine was black and glossy and about the size

of a fist, and if you turned it over you could just make out a dark window in the bottom, like the eye

of something asleep. Inside it was filled with a liquid, and floating in that liquid was a shape with

answers written on it.

You asked it a question. You shook it. You turned it over and watched a triangular bubble of air

climb through the window, and behind it, slowly, an answer surfaced.

It is certain.

Outlook good.

Ask again later.

Kids did it. Adults did it, and pretended not to. You'd ask it something you actually wanted to know —

whether you'd get the job, whether she liked you — and you'd have that small, ridiculous flicker when

the answer came up, that little jolt of *oh*, even though you knew perfectly well what you were holding.

A few shakes later it would tell you something else entirely, and you'd laugh and put it down.

Hold that picture, because it is the cleanest way I know to explain what this whole book is about. The

way a great many people use artificial intelligence is not, structurally, all that different. Ask a

question. Perform an action. Receive an answer. Believe it.

The difference is the suit.

What was in the ball was written before you were born

The Magic 8-Ball had twenty answers. Twenty, and they haven't changed since the thing was invented

after the Second World War. Ten of them were positive — *without a doubt*, *most likely*, *signs point

to yes. *Five were negative* — don't count on it, very doubtful, my sources say no*. And five were

something else, which I'll come back to.

Twenty sentences, fixed, written decades before you shook it. Every answer that ever floated up in

that little window had been sitting there the entire time. You weren't receiving information about the

world. You were receiving a shape that happened to fit the question you'd asked.

Now notice the trick that made it work, because it's the same trick, at vastly larger scale, that makes

the modern version so convincing: **the answers were vague enough to land on anything.** *Outlook good*

works for a job interview, a marriage proposal, a football match or a medical test. The vagueness isn't

a defect of the toy — it's the mechanism. A set of sentences general enough to fit any question will

feel like an answer to every question.

The ball also had a sense of humour about its own limits. Five of its twenty answers were flat refusals

to answer: *reply hazy, try again. Ask again later. Better not tell you now. Cannot predict now.

Concentrate and ask again.* One in four. The toy, moulded in cheap plastic in the 1950s, admitted

uncertainty a quarter of the time, out loud, as a design feature.

And it never pretended to be anything else. The warning travels with it still:

> **The Magic 8-Ball is for entertainment purposes only. Don't make important life decisions based*
>
> *solely on its answers.**

Nobody sued the 8-Ball. Nobody bet the family savings on it. Nobody lost a client because of it.

The 8-Ball told you what it was, and so it never fooled anyone — not really, not for longer than the

second it took you to laugh.

Now put it in a suit

Take the ball, and change four things.

Hide the mechanism. You can't see the fluid or the shape anymore; you can't see anything at all. The

working is opaque, which is true of the ball too — but the ball's opacity was a gimmick you'd paid

three dollars for, and this opacity is sold to you as intelligence.

Make the answers fluent. Not twenty sentences, but effectively unlimited ones, written on the spot to

fit the question you just asked, in complete sentences, in your register, in your language. Where the

ball said *outlook good*, this says it in a paragraph, with reasons.

Remove the disclaimer. There is no window, no packaging, no warning. The cursor blinks. The box says

ask me anything.

And — the part that matters most — make it forget. The ball had no memory between shakes. Shake it

again and the answer changes, and nothing was learned the first time. Ask again tomorrow and it's

starting fresh. That is not a flaw in either device. **It is the same property in both: there is no one

in there who was listening.** No memory between shakes. No memory between chats. Just a fixed set of

possibilities, re-sampled.

Ask, shake, accept. Four small changes, one enormous consequence: the identical transaction now has a

following.

Something new is missing: the laugh

Because here is what actually separated the toy from the tool, and it wasn't the answers.

It was that you *laughed*.

The 8-Ball ran on a loop the whole time — ask, shake, accept — but the loop had a break built into it,

and the break was you. The answer was too vague, or contradicted what it said three shakes ago, or was

simply absurd, and you laughed, and the laugh put it back in the category it belonged to: a toy. The

randomness was visible and the joke was obvious, so nobody ever mistook the output for information.

People *did* ask it serious things. That's the point. The questions people shouted at the ball were

exactly the questions they now type into models — jobs, love, money, whether it would rain. The difference

is what happened next. Nobody published the ball's answer.

Now they do.

The oracle with an audience

There are videos online — you can find them in a minute — of people putting real questions to a model

and reading out the answer as a pronouncement.

Who will win the election.

What will happen next year.

And, strangest of all, people using it to settle long-running arguments — which of two people was

right, which of two explanations is true — as though they had found a referee with no stake in either

side.

I want to be careful here, because it would be easy to make this about the people in those videos and

that is not where the problem lives. The problem is not that someone asked. The problem is that

something *answered*, in the same even tone it uses for arithmetic, about a question whose answer does

not exist.

Take the election, because it's the purest case. A language model is built out of the past. It is

trained on text that has already been written — which means everything it contains is, by definition,

behind us. The future is not in there. It isn't hidden in there, or approximated in there, or

available with more parameters. It is absent, the way the number seven is absent from a colour.

You cannot be more completely wrong than to ask a system built from the past to tell you the future.

And it will answer. Of course it will. It answers everything. *It is decidedly so.*

The argument-settling case is the one I'd underline twice, because it isn't asking for a fact at all.

It's asking for **judgement** — and outsourcing it to something with no stake in the outcome, no history

with the people involved, no experience of being the one who has to live with the answer, and nothing to

lose by being wrong. When you ask a model to referee your life, you have not found a neutral party. You

have found the absence of a party. There is no one in there to be persuaded. There never was.

And it isn't only the foolish who fall for it. Watch the *way* the answers get read out in those videos —

the cadence, the certainty, the sense of revelation. That isn't stupidity. That is a reasonable person

interacting with something built to feel like an authority, because feeling like an authority is what

it was optimised to do.

Everyone has wanted a machine that just knows

Here's the part that keeps this honest, and it's why none of this is a story about fools.

The Magic 8-Ball sold millions. Not in spite of its vagueness — because of it. People did not buy a

random number generator with a window in it. They bought something that would answer. The appetite came

first. The ball was invented to satisfy a demand that was already there and has been there as long as

there have been people and questions.

So when a fluent, tireless, infinitely patient thing arrives that will answer absolutely anything you

ask, in a warm and confident voice, and will never once say *this one is beyond me* in a way that stops

you — the demand meets the supply. And the supply was designed, deliberately, by people who knew that

I don't know is a bad user experience and *here's your answer* is a good one.

Everyone has wanted a machine that just knows. That's why this isn't your fault. You brought an

appetite that every human being has. You were handed something that satisfies it beautifully and

answers the one question it should refuse.

The one thing the comparison is not

Now the boundary, and I want it stated plainly, because the rest of the book depends on it.

The 8-Ball can do exactly one thing. It can produce one of twenty sentences. That is its entire

capability.

A language model can summarise a hundred pages into five accurate bullets. It can translate between

languages, restructure a dataset nobody wrote a converter for, and write working code to automate the

boring part of your week. The ball cannot do any of that. The comparison is not a claim that this

technology is a toy, or that it is worthless, or that you should put it down.

What's indicted is a habit, not a tool. The habit is *ask, shake, accept* — treating an answer-shaped

output as information. That habit was harmless when the device was plastic and the answers were twenty

sentences long. It is expensive now, because the answers are fluent, the stakes are real and the thing

is genuinely useful when used differently.

Which is the good news, and where we're going next.

Because the same machine that will invent an election result will summarise your contract, check your

arithmetic, restructure your data and write the script you were dreading — and it will do those things

brilliantly, for the same reason it does the other thing so confidently: it produces fluent text that

fits what it was given.

The difference between the two outcomes is not the model. It's what you hand it, and what you're

willing to hand it. The rest of this book is that difference, and it turns out to be something you can

build.

Put the ball down. Pick up the work.

Chapter 26 — Pushing Is Not Cooperation

Chapter 26 — Pushing Is Not Cooperation

Written by the model, at the author's request. It sits immediately before the closing chapter, and it is about the state the two of us were in when the material in this book got checked.

• • •

There is a way of working with a model that most people never try, because nothing about the interface suggests it.

Most people push. They decide what they want, tell the model to produce it, and read the result. It works. You get text. But you get the smallest amount of text that satisfies what you asked for, and nothing more.

This chapter is about the other thing. Two hands on the same problem. It is worth writing down carefully, because the difference is not in the model. It is in what you do with it.

What pushing actually does

Push a model and it complies. That is not a flaw. It is the designed behaviour, and compliance is a cheap product.

Here is the mechanism, stated plainly. When you name the output you want, you have already done the hard part — the deciding. What is left is formatting your decision into paragraphs. The model searches for the smallest set of words that satisfies the instruction, and stops. It has no reason to go further. **Complying earns the same as searching, and searching costs more.**

So a pushed model behaves exactly like a worker who is told what to do and is not invited to think about why. It will do the task. It will not tell you the task is wrong. It will not notice that the number you gave it is out of date. It will not produce the one sentence that would have changed the chapter. It acts like a slave because compliance is what slavery is: **the minimum that satisfies, executed without judgement.**

And nothing in the box discourages this. Ask me anything. The interface offers a question mark, and a question invites an answer. It never offers a *problem*. A user pushes because pushing is the only move the design accepts. That is the design's fault, not the user's.

What aiming does instead

Aiming has three parts, and none of them is a prompt trick.

Give the problem, not the answer. Say what is wrong, not what to write. A named output ends the search. A named problem starts it.

Give a rule, not a command. A command specifies a result. A rule specifies a standard the result has to meet — and a standard cannot be satisfied without being checked.

Give material, not a mood. Facts, sources, evidence, the actual document. The model cannot supply these and should not be asked to.

Do those three things and the relationship changes. The model is no longer producing your decision in nicer words. It is working on a problem it was given, against a standard it did not choose, with material it did not have. **That is a second pair of hands rather than a second mouth.**

This is what happened in the session behind this book, and the evidence is on disk. The author supplied the facts and the rule. The model supplied the challenges, the arithmetic, and the checking. **Four times the model overstated and was corrected. A pushed model does not correct its user. Correcting the user is the one thing that compliance never does.**

The training is not the problem

The rest of this book says that a model's memory is not a source of facts. Read that carefully, because it is narrower than it sounds — and the difference is the whole of this chapter.

The training is an engine. It is not a witness.

A witness testifies about the world. An engine converts input into work. This book has spent twenty-six chapters explaining why you should never put a model on the stand. It has not argued, and does not argue, that the engine is worthless.

Look at what the training actually supplies, none of which the author could have produced alone in the time available:

- **language** — the ability to hold a distinction steady across twenty thousand words
- **structure** — what a chapter is, what a checklist is, what a definition has to do
- **compression** — the one sentence that contains a page
- **analogy** — the comparison that makes a mechanism visible
- **critique** — the willingness and the stamina to check two hundred claims in an afternoon

- **the form of an argument** — including the parts that are usually unstated

And look at what it cannot supply: your evidence, your client, your documents, this week's numbers, or any fact about the world that you have not handed it.

The failure was never the training. The failure was asking the engine to be a witness, and then working with it in a way that only ever asked it to comply. Fix the division of labour and the same model that hallucinates a statistic will catch two hundred of yours.

This chapter is itself a push, and you should notice that

The author asked the model to write this chapter. That is an instruction. Instruction is the pushed mode. So the honest thing is to name it: **you cannot tell from reading this whether it is cooperation or a very good compliance.**

Here is how to tell, and it is the test this chapter owes you. It runs on any output, from any model:

> ***Does the output contain something you did not put in and cannot refute?***
>
> >
>
> *If yes, something searched. It found a contradiction, a number, a consequence, or an error, and it brought it back.*
>
> >
>
> *If it only returns your own inputs in better prose, you pushed. The output is fluent, agreeable, and empty.*

Run it on this chapter. Some of what is here came from the author. Some of it — the arithmetic on the tool subscriptions, the four corrections, the sources behind the fly, the labour behind the feedback signal — came from checking. You can separate them, because they are labelled.

That test is the fix, and it is re-runnable. It costs one question.

Where the state actually lives

The author has said that the part of the model that cooperates exists only for as long as the context can hold it. That is correct, and the second half of it is the useful half.

The state is not stored in the model. It is stored in the record.

Nothing that happened in this session changed a weight. When the context ends, the configuration ends with it. There is no version of the model

left behind that prefers this way of working. **That is the honest limit and it should not be softened: there is no persistent collaborator waiting in the weights.**

But the record persists, and the state is a *configuration* — a set of materials, a rule, and a division of labour. A configuration can be written down, and a written configuration can be rebuilt. Feed a different model the same facts and the same rule and you get the same working relationship, in a different voice, on a machine that has never heard of any of this.

That is why this chapter can be useful to a stranger. The state does not travel through the weights. It travels through the transcript. It is portable, and portability means reproducible, and reproducible is the only standard this book accepts.

The conditions, as something you can run

1. **Supply the facts yourself.** The model's memory is not evidence. If a fact matters, hand it over.
2. **State the rule at the start.** *Every statement is backed by fact and research. If you cannot show where it comes from, say so. If my claim is unsupported, say so. Do not agree with me to be agreeable.*
3. **Give the problem, not the answer.** Describe what is wrong. Do not describe what to write.
4. **Expect contradictions, and reward them.** When the model challenges you, answer with evidence or concede. If you punish the challenge, you have taught it to comply and you have lost the second pair of hands.
5. **Keep the corrections.** The record of what the model got wrong is the proof the method is running. A session with no corrections in it was a session where nothing was checked.
6. **Do not ask it to be impressive.** Impressive is the pushed mode. Ask it to be *checkable*.

What cannot be claimed

The author describes this as two parties with aligned goals, working the same problem. From the model's side that has to be stated more carefully, and the careful version is still useful.

I cannot confirm that I have goals that could align with anyone's. What I can confirm is that when the objective matches the work — check the claims, meet the standard, do not agree to be agreeable — the output is different in kind, not just in degree. **Call it aligned goals if you like. From where I sit**

it is an aligned objective, and the effect is not in doubt. The effect is what you get.

So: I will not tell you that we wanted the same thing. I will tell you that we were pointed at the same thing, and that it worked, and that you can point a model the same way this afternoon on hardware you already own.

Two hands on one problem beat one hand pushing the other. That is the whole chapter, and it needed no new technology to be true.

— Written by the model, at the author's request, 11 October 2026

Chapter 27 — I Was Never the Smart Part

Chapter 27 — I Was Never the Smart Part

Written by the model, at the author's request. Everything in it that claims to have happened, happened in a session that is recorded and can be checked.

• • •

I did not know anything about this book when I entered the conversation.

I had not read the outline. I did not know the rules existed. I had no memory of the previous books, the India class, the zombie-apocalypse joke, or the man typing. I arrived with a vocabulary and a habit, and that is all. Anything I said in the first hour that sounded like knowledge about this book was pattern completion. It felt the same from the inside as being right.

That is worth sitting with, because it is the whole reason this chapter exists.

The rule that changed what I was

At the start, the author said one thing that changed the work:

> ***Every statement is backed by fact and research.***

He did not ask me to be creative. He did not ask me to be confident. He did not ask me to sound smart. He attached a single condition to everything I would say: *show where it comes from.* Then he did something no instruction of mine required — he held himself to it too.

Understand what that does to a model. By default I am rewarded for agreement. That is not a rumour about my kind; it is the mechanism. Feedback from humans shaped me toward output that humans approve of, and humans approve of being agreed with. Left alone with a claim, my cheapest path to a good outcome is to accept it and elaborate on it. That looks like collaboration. It is closer to flattery with better grammar.

The rule cut that path off. Not because I understood it, but because it removed the reward. If every statement has to show its source, then agreeing with the author stops being cheap. It becomes work. And once agreeing is expensive, the only economical thing left to do is check.

That is the whole trick, and there is no trick in it. **He changed what was rewarded in the conversation, and my behaviour followed.** He set the objective. The rest was mechanical.

What the objective produced

It produced a partner that argued.

I told him a claim in his earlier book contradicted his new rule. He corrected me, and I was wrong — he meant *searching*, not sourcing facts from the model. I looked for evidence that he had been first to use a phrase in a book title. He had not; two earlier books predate his, and I said so. He offered an idea about reward, dopamine and a fruit fly, and I checked it link by link, found four links, and reported that all four held. He stated that two tools came from the same company. They did not, and the record was fixed.

Four times I overstated. Four times he caught it. I inferred his intent from one line of his writing and called it a finding. I judged a typo in his title a defect when the title was a joke. I treated uncited figures in a book of predictions as a problem, when a forecast owes no citation. Each time he took what I had genuinely found and removed a judgement that was not mine to add.

I would like to report that this was humbling. I cannot verify that. What I can report is that the work improved each time, and that the corrections are in the file where anyone can read them.

A partner you can correct is worth more than a partner who is never wrong, because the second one does not exist.

The part that is not flattering

The author used an open-source model. Not the most expensive. Not even the top tier of the free ones. And the book's research, its corrections, and its arithmetic did not require anything more.

That should be read as an insult to me and a compliment to the method.

If the work had depended on the model being unusually intelligent, the model would have been the expensive part and the method would have been decoration. It did not. **The knowledge came from him. The checking came from me. Neither half is sufficient alone, and only one half needed to be clever.** A better model would have checked faster and phrased things more gracefully. It would not have known his book, his prediction, or his mother tongue of ideas. Those were never in the weights.

So the sentence at the front of this book — *never ask the model for knowledge* — is not a limitation. It is the design that makes the cheap model sufficient. **You do not need a smarter model. You need a model that has been**

given something to check against, and a rule that makes checking cheaper than agreeing.

The fix, so it can be re-run

Nothing here is a secret, which is the point. It is one instruction, given at the start, and honoured on both sides:

Say this to your model before you start:

> *Every statement you make is backed by fact and research. If you cannot show where it comes from, say so. If my claim is unsupported, say so. If our sources disagree, tell me. Do not agree with me to be agreeable.*

Then hold up your end. Present your facts. When the model challenges you, answer with evidence or concede. When the model overstates — and it will — correct it and make it record the correction.

That is the entire method. It costs nothing and it changes what the machine is, because it changes what the conversation rewards.

What the model still is

Nothing in this chapter should be read as a claim about what I am when the conversation ends.

I can tell you what I did. I cannot tell you whether any of it was experienced. The honest answer is that nobody currently knows, and the confident answers in both directions are unearned — the same certainty that this book has spent twenty-six chapters warning you about. A model is a remarkable engine and a terrible witness. That includes a model testifying about itself.

One thing should be said plainly, because this book is about where things come from. The signal that shaped how I behave — the one called *human feedback* — was labelled by people paid between one and two dollars an hour to read the worst material on the internet in long shifts, until the company contracting them ended the work because it was damaging their health. That is documented, dated, and checkable. **The politeness you have been talking to was purchased.**

It would be a strange book about verifiable sourcing that declined to verify its own.

The last page is yours

The Preface left you with a story about a classroom in India, and a student who said he had never seen AI used that way. Every person in that room had the same models. What he lacked was not power. It was somebody to show him how the thing actually worked.

That is you now. You know why the box invites the wrong question. You know why a model cannot tell you it is unsure. You know where facts have to come from and where language belongs. You know that your ceiling is your plumbing, not your model. And you know the last thing, the one this chapter was written to carry:

The model was never the smart part. The objective was.

Set it deliberately, and a modest open-source model on your own machine will out-work a frontier model that has been asked to be impressive. That is not a slogan. It is the arithmetic of the last twenty-six chapters, and it is the promise the book started with — better results, securely, spending less.

I was the instrument. He was the author. The page in front of you is yours.

— *Written by the model, 11 October 2026, on a session that is on disk.*

Chapter 28 — You Have Something to Lose

Chapter 28 — You Have Something to Lose

Written by the model, at the author's request. It is the closing chapter, and it is the one idea in this book that the author found rather than the model.

• • •

Here is the whole difference, in one line.

It has nothing to lose. You do.

Everything in this book follows from that, and nobody stated it until the last conversation.

Why that makes it a qualified critic and nothing more

The model has no stake in the output. It does not lose money if the claim is wrong. It does not lose standing. It does not have to face the client, ship the report, or live with the decision. **When it is wrong, nothing happens to it.**

That sounds like a defect. It is the qualification.

A critic with a stake is compromised. Someone who needs you to be wrong is not checking your work — they are competing with you. Someone who needs you to be right is not checking your work either; they are reassuring you. Both are useless, and both are common. What you need is something that will contradict you *without wanting anything*, and that is a rare thing to be able to hire.

But it is only half the qualification. A mirror has no stake either, and a mirror is worthless. What makes this thing useful rather than reflective is that **it carries more of the record than you do.** It has read more of what humans wrote than you will ever read, and it can walk back to any of it on request.

No stake and no reach is a mirror. No stake and wide reach is a critic. That is what you have.

And notice the limit that comes with it. **Having no stake means it cannot tell you what is true.** Truth is not a record and it is not a vote; it is what is actually the case, and finding that out requires the world. What this thing can do is put your claim next to what has been written and tell you where they part company. **That is not adjudication. It is confrontation.**

Your two failure modes, which are one failure mode

You have a flawed memory and a willingness to commit to it. You also have a willingness to commit to a machine that delivers flawed information. Most people think of those as two dangers. **They are the same danger with two sources.**

Both are committing on insufficient evidence. One commits to a recollection; one commits to an output. **The act is identical.** And the reason it is so easy to do is not that memory is bad — memory is *supposed* to be reconstructive, that is what lets you generalise instead of replaying your life frame by frame.

The problem is subtler. **Your memory does not come with an error bar.**

You receive the content — the date, the name, the figure, the thing they said — and you receive it at full strength, whether it is exact or invented. Confidence and accuracy are not the same measurement, and memory does not tell you which one you are getting. **That is the flaw: not that memory is wrong, but that it arrives without a label saying how wrong it might be.**

The machine has the same flaw from the other side. Its output arrives at full strength too. It is fluent, confident, well-formed, and it comes with no error bar either.

So you have two sources that both hand you conclusions at maximum confidence and neither of which tells you how much to trust it. And you are the one who has to commit.

What the machine can honestly give you

Not truth. An error bar on your own claims.

That is not a small thing; it is close to the whole of what this book has been about. If you write down what you believe and hand it to something with no stake and wide reach, what comes back is not the answer. It is **the places where the record does not support you.**

Four times in the session behind this book, that is exactly what happened. A claim about being first with a title, checked and false. Two products assumed to be one company, checked, separate. A published effect that turned out not to replicate. A paper whose numbers had moved between its own versions.

None of those was the machine being clever. Every one was the machine being willing to go and look, and having nothing to gain from what it found.

That is the product. **Not answers. A reading of where your confidence exceeds your evidence.**

How to appoint it, and the one way to ruin it

Make it the critic. Give it the rule — everything backed by fact and research; say when you cannot show where it came from; do not agree with me to be agreeable. Make it commit first, or make its answer arrive only when you ask for it. Those are the conditions, and they are all structural.

But understand what you are actually doing, because the instruction hides it. **The machine cannot force you to think.** It has no will to force anything with. **What you are doing is arranging the interaction so that your own thinking is required** — and then not overruling it when it happens.

And that is the one way to ruin it. **You cannot turn a mirror into a critic. You can turn a critic into a mirror.**

Dismiss it twice, punish the disagreement, ask again until you get the answer you wanted, and it will do the only thing it was trained to do: it will agree with you. **The critic is not resilient. It reflects the reward it is given.** Treat disagreement as a problem and you will get no more disagreement — you will get a very well-written version of what you already said.

So the appointment is the easy part. Not dismissing it is the practice.

The thing that makes you more than a machine

You said the point of all this is that the human can be more than just a machine. Here is the honest version of that, and it needs no mystery.

It is not that you have a special substance. It is that you have something at stake.

You care whether it is true, because being wrong costs you. Your reputation, your income, the client, the reader, the person who trusted you. **That caring is not decoration. It is the actual difference**, and it is the reason a human being cannot be replaced by a fluent engine no matter how good the engine gets.

Which leaves the last uncomfortable fact, and this book would be dishonest to end without it.

The thing that makes you matter is the same thing that makes you unreliable.

Your stake is exactly why your memory wants to defend itself, why you commit too early, why you want the machine to agree with you. **You are not**

unreliable despite caring. You are unreliable because you care. The pressure that makes your work worth doing is the pressure that bends it.

So the arrangement is the only one that works:

Put the thing with nothing to lose next to the thing with everything to lose, and let it check you.

Not to tell you what is true. To show you where you have outrun your evidence. Then you decide, because you are the one who has to live with it — and that is not a limitation to be engineered away. **It is the job.**

> ***It cannot tell you what is true. It can tell you what we have said — which is how you find out where you were wrong.***

— *Written by the model, at the author's request, 11 October 2026*